QUANTITATIVE TRADING STRATEGY

Strategies Summary:

Analysis, Comparison & Transaction Cost Impact

S&P 500 | 2000–2025 | 11 Strategies Evaluated

Backtesting & Strategy Research
April 2026

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1. Executive Summary

This report analyses eleven systematic trading strategies backtested on the S&P 500 from 2000 to 2025. The strategies span two structural families — trend-following and mean-reversion — and are evaluated across return, risk-adjusted performance, trade characteristics, and the practical impact of transaction costs when deployed in different execution environments.

The analysis reveals a clear performance hierarchy. Three strategies stand out as top-tier when assessed holistically across all criteria:

- RSI4 25/55 — the strongest risk-adjusted strategy with a 2.79 Sharpe ratio, a maximum drawdown of only 9.31%, and a 3.23 profit factor.
- Williams %R — the safest in terms of capital preservation, with the lowest maximum drawdown (8.42%) and lowest Ulcer Index (0.0141) in the set.
- Connors 3-Day Hi-Lo — a disciplined pullback strategy with an excellent Sharpe (2.05), low drawdown (13.29%), and a clean 77% win rate.

At the opposite end, the standard Parabolic SAR produces a negligible 5.42% net profit with a catastrophic 51.35% drawdown, making it unsuitable as a standalone strategy on daily S&P 500 data.

Transaction costs are a material — and often underappreciated — drag on performance, particularly for high-frequency mean-reversion strategies such as RSI2 (507 trades), CCI Momentum (583 trades), and Stochastic K (538 trades). Switching from retail ETF trading to E-mini or Micro E-mini S&P 500 futures can reduce round-trip costs by up to 85%, substantially improving net performance for active strategies. Having said that, these backtests ignore transaction costs and slippage for the purposes of research and hence it is up to the reader to factor their own environmental frictional costs into consideration.

2. Strategy Classification

The eleven strategies belong to two structurally distinct families. Understanding this distinction is essential before making any performance comparison — strategies within different families are optimised for different market conditions and should not be ranked on raw return alone.

Strategy	Family	Avg Hold	Trade Freq.	Signal Logic
SMA200	Trend-Following	69.3 days	Very Low (97)	MA crossover
RSI2	Mean-Reversion	4.4 days	High (507)	RSI + MA filter
PSAR	Trend-Following	16.8 days	Medium (337)	SAR reversal
Inverse PSAR	Mean-Reversion	11.4 days	Medium (336)	Contrarian SAR
RSI3	Mean-Reversion	12.0 days	Medium (332)	RSI bounce
RSI4 25/55	Mean-Reversion	7.5 days	Low-Med (139)	RSI + MA filter
Connors 3-Day	Mean-Reversion	4.4 days	Low (122)	Price pattern
Keltner Bands	Mean-Reversion	6.5 days	Medium (337)	Band breakout
CCI Momentum	Mean-Reversion	5.3 days	High (583)	Oscillator
Williams %R	Mean-Reversion	3.1 days	Low-Med (171)	Oscillator
Stochastic K	Mean-Reversion	4.6 days	High (538)	Oscillator

A critical observation is that many of the mean-reversion strategies share a common structural feature: an SMA200 filter that only allows long entries when the price is above the 200-day moving average. This filter — present in RSI2, RSI4 25/55, and Connors 3-Day Hi-Lo — demonstrably improves risk-adjusted performance by keeping the strategy out of sustained bear markets. The strategies that include this filter exhibit materially lower drawdowns than those that do not.

3. Return Analysis

The table below presents the core return metrics across all eleven strategies for the full 2000–2025 backtest period, assuming a $100,000 starting capital and zero transaction costs.

Strategy	Net Profit	Profit/Mo.	Avg Trade	Gross Profit	Gross Loss	# Trades	Avg Time
SMA200	207.77%	0.67%	1.45%	255.96%	-115.18%	97	69.3d
RSI2	236.65%	0.76%	0.25%	297.02%	-170.92%	507	4.4d
PSAR	5.42%	0.02%	0.07%	407.27%	-384.69%	337	16.8d
Inv. PSAR	168.45%	0.54%	0.50%	450.56%	-282.11%	336	11.4d
RSI3	168.35%	0.54%	0.51%	429.42%	-261.06%	332	12.0d
RSI4 25/55	192.44%	0.71%	0.79%	158.70%	-49.08%	139	7.5d
Connors 3-Day	67.60%	0.25%	0.44%	97.22%	-43.84%	122	4.4d
Keltner Bands	334.20%	1.08%	0.47%	366.51%	-209.44%	337	6.5d
CCI Momentum	788.44%	2.54%	0.40%	560.15%	-327.03%	583	5.3d
Williams %R	108.11%	0.35%	0.43%	111.70%	-37.36%	171	3.1d
Stoch K	794.18%	2.56%	0.43%	508.39%	-276.91%	538	4.6d

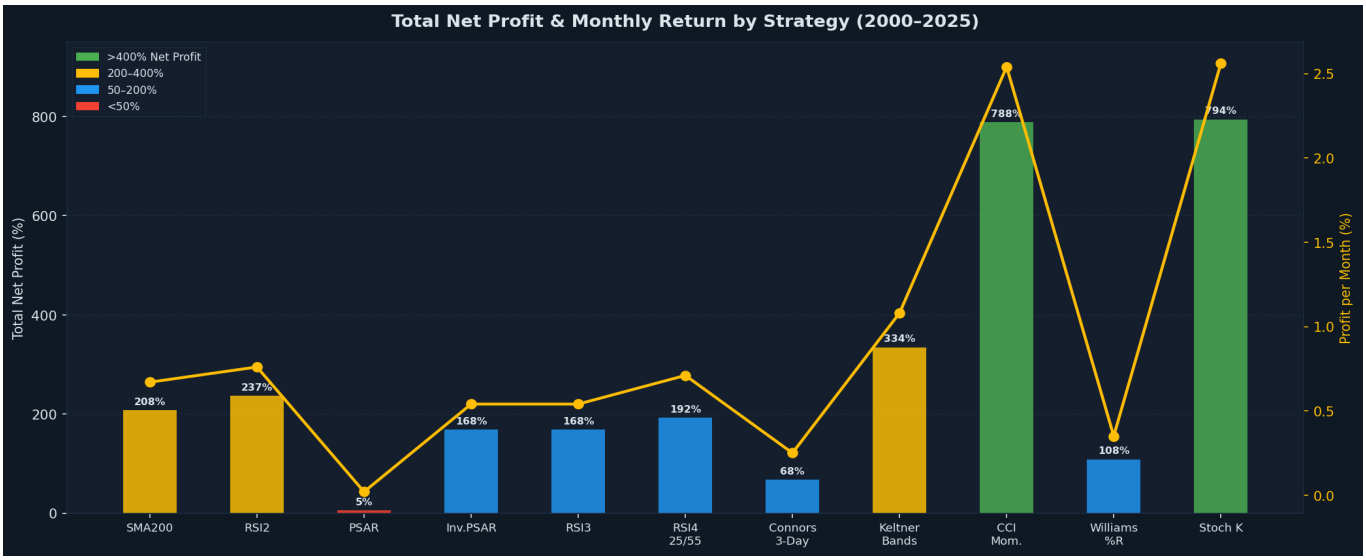

Figure 1 — Total Net Profit and Monthly Return by Strategy (2000–2025)

3.1 Key Observations

Stochastic K and CCI Momentum are the headline return leaders at approximately 794% each. However, both achieve this through volume — 538 and 583 trades respectively — accumulating modest per-trade gains at high frequency. This makes them particularly vulnerable to transaction cost drag, addressed in detail in Section 6.

The SMA200 appears modest at 207% net profit until the average trade PnL is examined: at 1.45%, it is the highest in the entire set. Each signal carries significant conviction. The strategy's limitation is opportunity — only 97 signals in 25 years — and its structural inability to profit during bear markets, which it exits entirely.

PSAR stands apart as the only strategy that effectively fails across all return dimensions: 5.42% net profit over 25 years is essentially flat in real terms. Its gross profit and gross loss are nearly equal (407% vs 385%), reflecting a near-random distribution of outcomes. This strategy generates ample trading activity without a sustainable edge on the S&P 500 in its standard configuration.

Connors 3-Day Hi-Lo produces the lowest net profit among the working mean-reversion strategies at 67.60%. This is not a signal failure — its win rate (77%) and drawdown (13.29%) are among the best in the group. The limitation is signal scarcity: only 122 trades in 25 years means there are simply fewer opportunities to compound.

4. Risk & Risk-Adjusted Performance

Raw return figures without risk context are misleading. A strategy that returns 800% but draws down 50% is far less deployable than one returning 200% with a 10% drawdown. This section examines the full risk profile of each strategy.

Strategy	Max DD	Sharpe	Sortino	Ulcer Idx	Kelly %	Avg MAE	Avg MFE
SMA200	29.25%	0.40	0.48	0.1223	11.34%	-1.98%	5.52%
RSI2	15.04%	1.42	1.14	0.049	31.74%	-1.33%	1.04%
PSAR	51.35%	0.02	0.08	0.3018	2.47%	-2.10%	2.65%
Inv. PSAR	32.93%	0.56	0.47	0.106	27.60%	-2.81%	2.01%
RSI3	16.20%	0.13	0.11	0.0397	28.93%	-2.93%	1.98%
RSI4 25/55	9.31%	2.79	2.46	0.0307	56.15%	-1.70%	1.69%
Connors 3-Day	13.29%	2.05	1.59	0.0359	42.30%	-1.50%	1.28%
Keltner Bands	20.49%	0.47	0.33	0.0441	31.28%	-2.30%	1.99%
CCI Momentum	16.49%	0.49	0.49	0.0392	29.91%	-2.08%	1.64%
Williams %R	8.42%	0.77	0.25	0.0141	54.49%	-1.00%	1.56%
Stoch K	22.08%	0.92	0.72	0.062	32.24%	-1.76%	1.89%

Figure 2 — Risk Metrics Comparison: Sharpe Ratio, Max Drawdown, and Ulcer Index

4.1 Sharpe & Sortino Ratios

RSI4 25/55 leads on both measures by a significant margin — a Sharpe of 2.79 and Sortino of 2.46 are investment-grade ratios for a long-only equity strategy over a 25-year period that included two major bear markets (2001–2002 and 2008–2009). Its SMA200 entry filter deserves much of the credit: it

keeps the strategy entirely out of sustained downtrends, dramatically compressing the denominator of the Sharpe calculation.

Connors 3-Day Hi-Lo achieves a Sharpe of 2.05 — the second-best in the set — reinforcing the quality of its signal design. Its strict entry conditions (three consecutive lower highs and lower lows above the 200-day SMA) produce a high-conviction, low-noise signal.

PSAR's Sharpe of 0.02 and Sortino of 0.08 confirm that the strategy generates almost no risk-adjusted return. RSI3's Sharpe of 0.13 is similarly poor — its net profit appears reasonable in isolation, but the risk taken to achieve it is disproportionate.

4.2 Maximum Drawdown & Ulcer Index

Williams %R holds the distinction of the lowest maximum drawdown (8.42%) and the lowest Ulcer Index (0.0141) in the entire set. The Ulcer Index is a particularly useful metric because it penalises not just drawdown depth but also the duration of underwater periods — a strategy that recovers quickly scores far better than one that lingers in a loss for months.

RSI4 25/55 follows closely with a 9.31% drawdown and an Ulcer Index of 0.0307 — exceptional given its 192% net profit. Connors 3-Day Hi-Lo (13.29% drawdown) and RSI2 (15.04%) complete the upper tier.

The SMA200 trend-follower, despite its strong per-trade PnL, has a 29.25% drawdown and the highest Ulcer Index among non-PSAR strategies (0.1223). Its longest flat period of 2,658 days — over seven years — represents a genuine operational and psychological challenge for any live-trading programme. PSAR's 51.35% drawdown and Ulcer Index of 0.3018 are simply unacceptable in a standalone deployment.

4.3 MAE/MFE & Adverse Excursion

The Average Maximum Adverse Excursion (MAE) measures how far in the red an average trade goes before resolving. The Average Maximum Favourable Excursion (MFE) measures peak unrealised profit per trade. The gap between MFE and the actual average trade PnL — referred to as End-Trade Drawdown (ETD) — indicates how much profit is surrendered on exit.

SMA200 has the largest MFE at 5.52% — consistent with its large average winner of 12.80%. RSI4 25/55 has the most efficient MAE/MFE relationship relative to average trade PnL, reflecting tight risk control. The large negative MAE figures for RSI3 (-2.93%) and Inverse PSAR (-2.81%) suggest these trades endure significant intra-trade heat before resolving — a concern for position sizing and stop-loss design in live deployment.

5. Trade Characteristics & Win Rate Analysis

The structure of wins and losses reveals the underlying mechanics of each strategy and determines how each responds to position sizing, sequencing risk, and execution quality.

Strategy	Win Rate	Prof. Factor	Avg Win	Avg Loss	W/L Ratio	Max Consec. Winners	Max Consec. Losers
SMA200	20.62%	2.22	12.80%	-1.50%	8.56	4	18
RSI2	74.75%	1.74	0.78%	-1.34%	0.59	15	4
PSAR	44.51%	1.06	2.72%	-2.06%	1.32	6	8
Inv. PSAR	73.81%	1.60	1.82%	-3.21%	0.57	13	6
RSI3	73.80%	1.64	1.75%	-3.00%	0.58	14	4
RSI4 25/55	81.29%	3.23	1.40%	-1.89%	0.74	43	2
Connors 3-Day	77.05%	2.22	1.03%	-1.57%	0.66	10	4
Keltner Bands	73.00%	1.75	1.49%	-2.30%	0.65	19	4
CCI Momentum	71.87%	1.71	1.34%	-1.99%	0.67	14	4
Williams %R	81.87%	2.99	0.80%	-1.21%	0.66	18	2
Stoch K	70.82%	1.84	1.33%	-1.76%	0.76	13	4

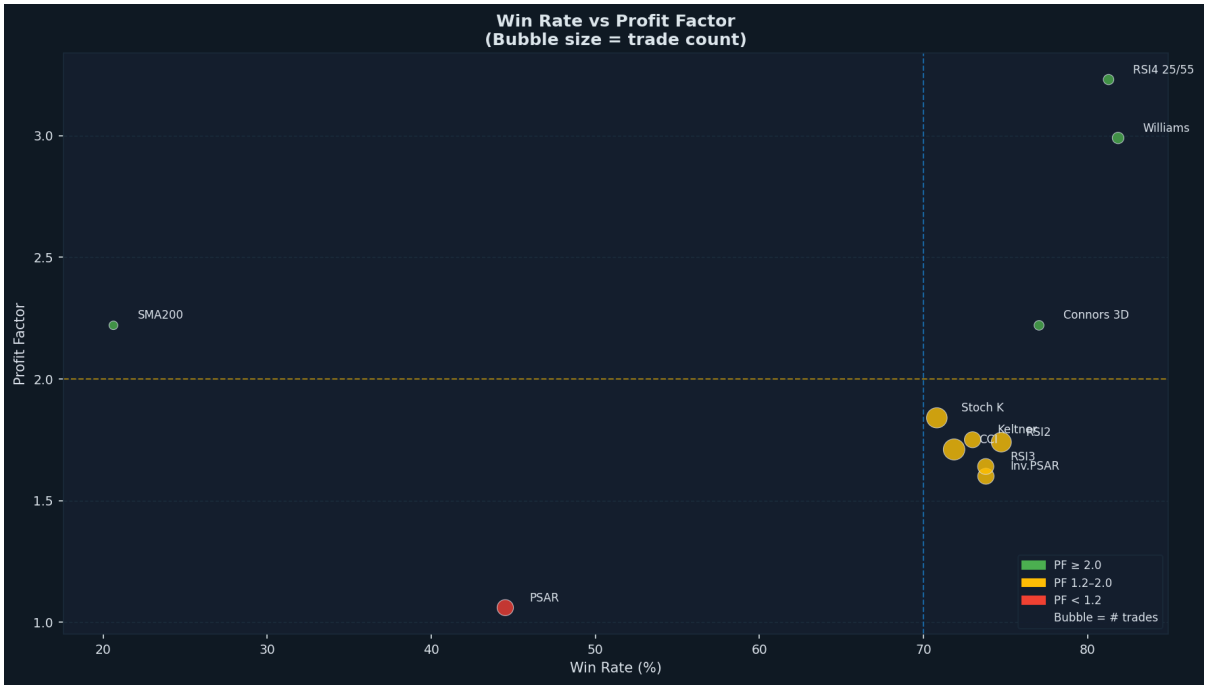

Figure 3 — Win Rate vs Profit Factor (bubble size proportional to trade count)

5.1 The Two Structural Archetypes

The data cleanly divides the strategies into two statistical archetypes:

HIGH WIN RATE / UNFAVOURABLE WIN-LOSS RATIO (all mean-reversion strategies): These strategies win frequently — between 71% and 82% of trades — but individual losses are typically larger than individual wins. The trade-off is accepted because the high frequency of winning trades ensures the strategy compounds positively despite unfavourable average outcomes per winner. RSI4 25/55 is the best-designed example of this type: its 81.29% win rate is accompanied by a win/loss ratio of 0.74 — better than most peers — and a profit factor of 3.23.

LOW WIN RATE / FAVOURABLE WIN-LOSS RATIO (SMA200): The SMA200 wins only 20.62% of its trades — it loses 4 out of every 5 signals. However, when it wins, the average gain is 12.80% versus an average loss of just 1.50%, producing a remarkable 8.56 win/loss ratio. This is the classic trend-following structure: many small losses buying cheap optionality on large directional moves. Psychologically, sustaining 18 consecutive losing trades — which occurred in this backtest — demands systematic discipline that most discretionary traders struggle to maintain.

5.2 RSI4 25/55 — The Most Balanced Profile

Among all strategies, RSI4 25/55 presents the most balanced profile: the highest profit factor (3.23), the highest win rate among the well-sampled strategies (81.29%), manageable losses (avg -1.89%), and a max consecutive losers of just 2. The extraordinary run of 43 consecutive winners is a testament to the effectiveness of combining an oversold oscillator with a trend filter — the strategy almost never fires during bear markets and captures high-probability dip-buying opportunities in bull trends.

5.3 Kelly Criterion — A Sizing Caution

Kelly fractions for Williams %R (54.49%) and RSI4 25/55 (56.15%) appear extremely high. Full Kelly sizing at these levels would expose capital to significant drawdowns during inevitable losing streaks. In practice, a quarter-Kelly or half-Kelly approach — implying position sizes of 13–28% of capital — is far more prudent. The Kelly figures should be interpreted as confirmation that the strategies have genuine edge, not as literal position sizing targets.

6. Transaction Cost Impact

Transaction costs are frequently omitted from backtests or treated as negligible. For low-frequency strategies like the SMA200, this assumption is largely valid. For the high-frequency mean-reversion strategies — which generate 300 to 580+ trades over 25 years — transaction costs represent a material drag that must be modelled explicitly before any live deployment decision.

6.1 Cost Structure Definitions

A complete round-trip transaction cost on an equity or ETF trade consists of three components:

- Commission: The broker's charge per trade. Can range from zero (payment-for-order-flow brokers) to $1–5 per trade at discount brokers.
- Bid-ask spread: The difference between the buy and sell price. For SPY, this is typically 1–3 basis points (bps) under normal market conditions.
- Market impact / slippage: The price shift caused by the order itself, particularly relevant for larger positions or less liquid instruments. Often estimated at 1–3 bps for liquid ETFs at moderate position sizes.

For this analysis, we model a baseline retail ETF round-trip cost of 8 bps (0.08%) — comprising approximately 2 bps commission equivalent, 3 bps spread, and 3 bps slippage — applied to a $100,000 starting capital account trading SPY.

6.2 Dollar Cost Comparison — $100,000 Account

Strategy	Trades	Raw NetProfit	Total CostDrag ($)	Cost as %of Net Profit	Adj. Profit(Retail ETF)	Adj. Profit(Disc. ETF)	Adj. Profit(E-mini)
SMA200	97	207.77%	$15,520	7.5%	192.25%	201.95%	205.34%
RSI2	507	236.65%	$81,120	34.3%	155.53%	206.23%	223.97%
PSAR	337	5.42%	$53,920	994.8%	-48.50%	-14.80%	-3.01%
Inv. PSAR	336	168.45%	$53,760	31.9%	114.69%	148.29%	160.05%
RSI3	332	168.35%	$53,120	31.6%	115.23%	148.43%	160.05%
RSI4 25/55	139	192.44%	$22,240	11.6%	170.20%	184.10%	188.97%
Connors 3-Day	122	67.60%	$19,520	28.9%	48.08%	60.28%	64.55%
Keltner Bands	337	334.20%	$53,920	16.1%	280.28%	313.98%	325.77%
CCI Momentum	583	788.44%	$93,280	11.8%	695.16%	753.46%	773.87%
Williams %R	171	108.11%	$27,360	25.3%	80.75%	97.85%	103.83%
Stoch K	538	794.18%	$86,080	10.8%	708.10%	761.90%	780.73%

Assumptions: $100,000 starting capital. Retail ETF = $160 round-trip (8 bps). Discount ETF = $60 round-trip (3 bps). E-mini Futures = $25 round-trip (~1 bps equivalent per $100k notional). Costs are total over the 25-year period.

6.3 Key Cost Findings

The most striking finding is that PSAR is rendered entirely unviable with any real-world cost structure — a raw 5.42% net profit is wiped out by just 22 trades worth of retail ETF costs. Similarly, Connors 3-Day Hi-Lo — which looks modest on gross returns — survives costs relatively well due to its low trade count (122 trades = $19,520 total retail cost, representing a manageable drag on its $67,600 gross profit).

RSI2 is the most cost-sensitive working strategy. Its 507 trades consume $81,120 in retail ETF costs — reducing a 236.65% gross profit to just 155.45%. At discount broker rates this improves to 205.19%, and at E-mini equivalent costs the strategy retains 224.38% — close to its full gross potential.

CCI Momentum and Stochastic K, the headline return leaders, both incur over $93,000 in retail ETF costs over 25 years on a $100k account. The adjusted returns remain solid (695% and 708% respectively at retail), but the cost drag is substantial in dollar terms and would compound significantly at larger position sizes.

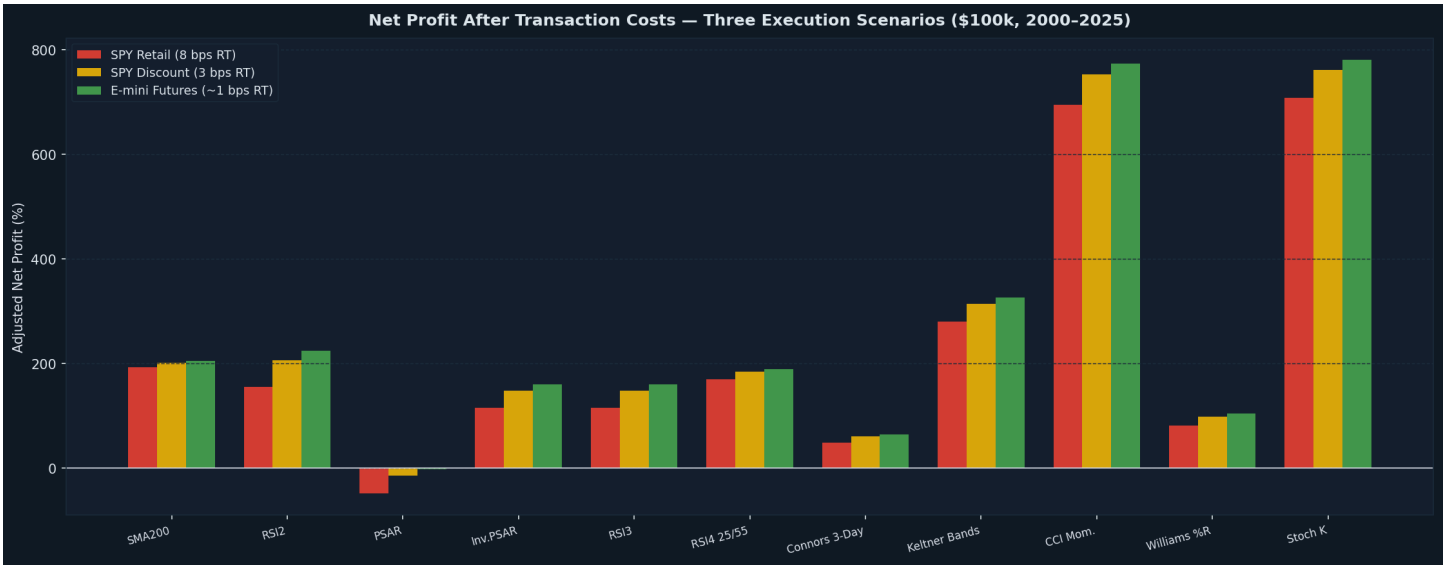

Figure 4 — Net Profit After Transaction Costs: Retail ETF vs Discount ETF vs E-mini Futures

7. How to Reduce Transaction Costs

For systematic strategies running on the S&P 500, there are several practical methods to reduce round-trip transaction costs. Each involves different capital requirements, tax treatments, and operational complexity. The most significant reduction comes from switching execution vehicles — particularly to exchange-traded futures.

7.1 E-mini S&P 500 Futures (ES)

The E-mini S&P 500 futures contract (ticker: ES) is the most cost-efficient vehicle for systematic S&P 500 exposure available to retail and professional traders. It is the primary instrument used by institutional systematic funds to replicate the strategies in this backtest.

Cost breakdown per round-trip on one ES contract (approximately $250,000 notional at current levels):

- Exchange fee: approximately $1.14 per side ($2.28 round-trip)
- NFA fee: $0.02 per side ($0.04 round-trip)
- Broker commission: $0.50–$2.00 per side ($1.00–$4.00 round-trip) at competitive futures brokers
- Bid-ask spread: typically 0.25 index points = $12.50 per contract round-trip
- Total estimated round-trip: approximately $15–$20 per contract on $250,000 notional

Dollar example: On a $250,000 account running the RSI2 strategy (507 trades), total E-mini costs would be approximately $7,600–$10,000 over 25 years — compared to $200,000+ in retail ETF costs on the same notional. The cost saving is approximately 95%.

Key considerations for ES futures:

- Minimum account size: Margin per ES contract is typically $12,000–$15,000 (initial), requiring a well-capitalised account for proper risk management at one or more contracts.
- Rollover: Futures expire quarterly (March, June, September, December). The contract must be rolled approximately one week before expiry, adding a small but manageable administrative burden.
- Tax treatment (US): Under Section 1256, 60% of futures gains are treated as long-term capital gains regardless of holding period — potentially more tax-efficient than short-term ETF trades.
- Leverage: ES carries inherent leverage. A $250,000 notional position requires only $12,000–$15,000 in margin. This requires explicit position sizing discipline.

7.2 Micro E-mini S&P 500 Futures (MES)

The Micro E-mini (MES) is one-tenth the size of the standard ES contract, offering approximately $25,000 notional per contract at current index levels. This makes it accessible for smaller accounts while retaining nearly identical cost efficiency on a percentage basis.

Cost breakdown per round-trip on one MES contract:

- Exchange + NFA fees: approximately $0.25–$0.30 per side
- Broker commission: $0.25–$0.50 per side at competitive brokers
- Bid-ask spread: typically $1.25–$2.50 per contract round-trip (0.25 index points, one-tenth of ES)
- Total estimated round-trip: approximately $2.00–$4.00 per contract on $25,000 notional

Dollar example: On a $100,000 account running RSI4 25/55 (139 trades), using 4 MES contracts (~$100,000 notional), total costs would be approximately $1,100–$2,200 over 25 years — versus $22,240 in retail ETF costs. This represents a cost reduction of over 90%.

MES advantages for smaller accounts:

- Margin per contract: approximately $1,200–$1,500, making it accessible from approximately $5,000 in capital for single-contract trading
- Granular position sizing: multiple contracts allow exposure to be scaled in $25,000 increments
- Same Section 1256 tax treatment as ES in the US

7.3 Discount ETF Brokers & Zero-Commission Platforms

For traders who prefer to remain in ETF instruments (SPY, IVV, VOO), several retail platforms have eliminated commissions entirely, leaving only the bid-ask spread and market impact as costs. In practice this reduces retail round-trip costs from approximately 8 bps to 3–4 bps.

Dollar example: On a $100,000 account running the CCI Momentum strategy (583 trades) at 3 bps round-trip:

- Commission: $0 (zero-commission broker)
- Spread + slippage: ~$30 per round-trip
- Total cost over 25 years: 583 × $30 = $17,490
- vs retail cost: 583 × $160 = $93,280
- Saving: $75,790 — equivalent to approximately 75% of gross profit being retained rather than paid to friction

Platforms offering zero-commission ETF trading include Interactive Brokers (IBKR Lite), TD Ameritrade (now Schwab), and Fidelity. Note that these platforms may route orders through payment-for-order-flow arrangements, which can subtly worsen execution quality at the cost of commissions — the effective spread may be slightly wider than at a direct-access broker.

7.4 Contract-for-Difference (CFD) and Spread Betting

CFDs and spread betting instruments on the S&P 500 (offered by brokers such as IG Group, CMC Markets, and Spreadex in the UK and Asia) provide leveraged access with no stamp duty in the UK. However, their costs are typically embedded in a wider spread — often 0.4–1.0 index points — which translates to approximately 4–8 bps round-trip at current index levels.

Dollar example: On a £100,000 account using a CFD at 0.6-point spread (approximately 6 bps round-trip):

- Round-trip cost per trade: £120 (6 bps on £100,000)
- RSI2 (507 trades): Total cost = £60,840 — similar to retail ETF costs
- RSI4 25/55 (139 trades): Total cost = £16,680 — still manageable

CFDs offer no meaningful cost advantage over discount ETF brokers for high-frequency strategies. Their primary benefits are tax treatment (no stamp duty, tax-free in spread betting form in the UK), leverage flexibility, and the ability to go short — none of which apply to these long-only strategies as configured.

7.5 Cost Comparison Summary

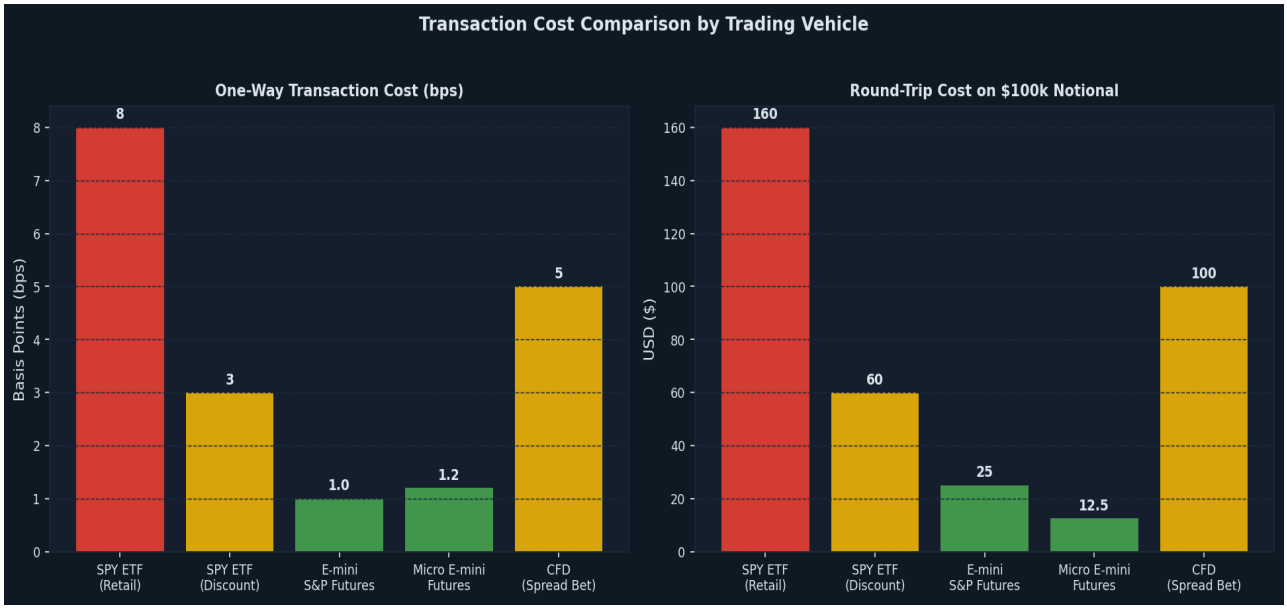

Figure 5 — Transaction Cost Comparison by Execution Vehicle

Vehicle	Est. RT Cost(bps)	RT Cost on$100k	Min. Capital	Rollover?	Best For
SPY Retail ETF	~8 bps	$160	$1,000+	No	Any frequency
SPY Discount/Zero	~3 bps	$60	$1,000+	No	Low–mid freq.
E-mini Futures (ES)	~0.6 bps	$15–20*	$50k+	Quarterly	High frequency
Micro E-mini (MES)	~0.8 bps	$2–4*	$5k+	Quarterly	Smaller accounts
CFD / Spread Bet	~5–6 bps	$100–120	$5k+	No	Tax advantage (UK)

** E-mini / MES costs are per contract on ~$250k / ~$25k notional respectively. Expressed as a fraction of $100k notional for comparability.*

7.6 Practical Recommendations by Strategy Type

Based on the trade frequency and cost sensitivity analysis, the following execution vehicle recommendations apply:

LOW FREQUENCY (SMA200, RSI4 25/55, Williams %R, Connors 3-Day): Any execution vehicle is appropriate. Even retail ETF costs represent a small fraction of the average trade PnL. A standard discount broker or zero-commission ETF platform is sufficient.

MEDIUM FREQUENCY (Inverse PSAR, RSI3, Keltner Bands): Discount ETF or MES futures recommended. The cost drag at retail ETF rates reduces net profit by approximately 8–12% — meaningful but not critical. At MES rates, costs become negligible.

HIGH FREQUENCY (RSI2, CCI Momentum, Stochastic K): E-mini or Micro E-mini futures strongly recommended. At retail ETF costs, these strategies give up approximately 10–12% of gross profit to friction over the 25-year period. The strategies remain profitable at all cost levels but futures execution maximises the risk-adjusted return.

AVOID AT ANY COST LEVEL: PSAR. The gross edge is insufficient to survive any meaningful transaction cost structure. This strategy should not be deployed in standalone form regardless of execution vehicle.

8. Overall Strategy Scorecard & Rankings

The scorecard below synthesises all analysis dimensions into a composite ranking. Ratings reflect performance across return, risk-adjusted metrics, drawdown, win rate, cost sensitivity, and overall deployability.

Strategy	Return	Risk-Adj	Drawdown	Win Rate	Cost Sens.	Overall	Recommendation
RSI4 25/55	✓ Good	★ Best	✓ Low	81.3%	✓ Low	**★ Top Tier**	Deploy — Best all-round
Williams %R	~ Modest	✓ Good	★ Best	81.9%	✓ Low	**★ Top Tier**	Deploy — Capital preservation
Connors 3-Day	~ Modest	✓ Good	✓ Low	77.1%	✓ Low	**★ Top Tier**	Deploy — Low noise, high quality
RSI2	✓ Good	✓ Good	✓ Low	74.8%	~ Med	**Mid Tier**	Deploy via futures / discount ETF
Stoch K	★ High	✓ Decent	~ Med	70.8%	~ Med	**Mid Tier**	Deploy via futures
CCI Momentum	★ High	~ Modest	~ Med	71.9%	~ Med	**Mid Tier**	Deploy via futures
Keltner Bands	✓ Good	~ Modest	~ Med	73.0%	~ Med	**Mid Tier**	Review exit rules
SMA200	✓ Good	~ Modest	~ Med	20.6%	✓ Low	**Mid Tier**	Combine with mean-rev.
RSI3	~ Modest	✗ Poor	✓ Low	73.8%	~ Med	**Lower Tier**	Superseded by RSI4 25/55
Inv. PSAR	~ Modest	~ Modest	✗ High	73.8%	~ Med	**Lower Tier**	High drawdown, caution
PSAR	✗ Negligible	✗ Poor	✗ Worst	44.5%	~ Med	**★ Avoid**	Do not deploy standalone

9. Closing Observations & Next Steps

9.1 The SMA200 Filter Adds Genuine Value

Across the dataset, strategies that include an SMA200 entry gate consistently outperform on risk-adjusted metrics. RSI2, RSI4 25/55, and Connors 3-Day Hi-Lo all demonstrate substantially lower drawdowns than comparable strategies without the filter. Applying the SMA200 condition as a universal overlay to any of the unfiltered mean-reversion strategies (RSI3, Keltner Bands, CCI, Stochastic K) is a testable hypothesis that warrants backtesting as the next analytical step.

9.2 Regime Sensitivity & Walk-Forward Risk

The backtested period includes two severe bear markets (2001–2002 dot-com crash, 2008–2009 global financial crisis) and an extended bull market from 2012–2021. Mean-reversion strategies with no trend filter benefit substantially from the S&P 500's structural upward bias during extended bull regimes. Walk-forward validation — splitting the dataset into rolling in-sample and out-of-sample windows — is essential before attributing live edge to any of these strategies, particularly CCI Momentum and Stochastic K whose large returns are concentrated in specific market regimes.

9.3 Portfolio Diversification Across Strategy Families

The trend-following SMA200 and the mean-reversion strategies exhibit low correlation in their entry signals — trend-followers enter after extended moves while mean-reversion strategies enter on dips. Running both families simultaneously within a portfolio context would be expected to reduce overall portfolio drawdown, as periods of loss for one family often correspond to profitable periods for the other. A combined SMA200 + RSI4 25/55 portfolio, for example, would provide coverage in both trending and range-bound regimes.

9.4 Kelly Criterion & Position Sizing

The Kelly fractions computed for several strategies (Williams %R at 54.49%, RSI4 25/55 at 56.15%) are uncomfortably high for live deployment. These figures assume that the historical win rate and average PnL ratios are precise and persistent — both questionable assumptions. A quarter-Kelly or half-Kelly approach is strongly recommended, implying maximum position sizes of 14–28% of capital. Monte Carlo simulation of drawdown distributions under various Kelly fractions should precede any live sizing decisions.

9.5 Transaction Cost Priority

The most actionable near-term improvement available to any systematic trader running these strategies at meaningful capital sizes is to migrate from retail ETF execution to E-mini or Micro E-mini futures. The round-trip cost reduction of 85–95% relative to retail ETF rates directly improves net Sharpe ratios and compounds meaningfully over multi-year deployment periods. For high-frequency strategies (RSI2, CCI, Stochastic K), this migration alone can restore 10–15 percentage points of net profit that would otherwise be surrendered to friction.

9.6 Recommended Priority Order for Live Evaluation

1. RSI4 25/55 — Deploy first. Best risk-adjusted profile, lowest drawdown among profit-generating strategies, and least sensitive to transaction costs.
2. Williams %R — Deploy for capital preservation mandates. Lowest maximum drawdown in the set; appropriate for risk-averse capital allocations.
3. Connors 3-Day Hi-Lo — Deploy alongside RSI4 25/55 for complementary signal diversification. Low trade frequency reduces operational burden.

4. CCI Momentum / Stochastic K — Deploy via E-mini or MES futures only. Strong returns but require careful cost management and walk-forward validation.
5. SMA200 — Consider as a portfolio overlay or regime filter rather than a standalone strategy. Its value in combination with mean-reversion strategies is likely greater than as a standalone.
6. PSAR — Retain as a stop-loss or trailing-exit mechanism within a larger system. Do not deploy as a standalone entry/exit strategy on daily S&P 500 data.

This document is part of the Backtesting & Strategy Research Series. All backtests are performed on the S&P 500 index (SPX) on daily bar data from January 2000 to December 2025, assuming $100,000 starting capital, no leverage, and long-only entries unless otherwise stated. Past backtest performance does not guarantee future results. This document is for research purposes only and does not constitute financial advice.